```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df=pd.read_csv("AirQuality2.csv",sep=';')

df.head()
```

```
         Date      Time CO(GT)  PT08.S1(CO)  NMHC(GT) C6H6(GT)
PT08.S2(NMHC)  \
0  10/03/2004  18.00.00    2,6       1360.0     150.0     11,9
1046.0
1  10/03/2004  19.00.00      2       1292.0     112.0      9,4
955.0
2  10/03/2004  20.00.00    2,2       1402.0      88.0      9,0
939.0
3  10/03/2004  21.00.00    2,2       1376.0      80.0      9,2
948.0
4  10/03/2004  22.00.00    1,6       1272.0      51.0      6,5
836.0

   NOx(GT)  PT08.S3(NOx)  NO2(GT)  PT08.S4(NO2)  PT08.S5(O3)     T
RH  \
0    166.0        1056.0    113.0        1692.0       1268.0  13,6
48,9
1    103.0        1174.0     92.0        1559.0        972.0  13,3
47,7
2    131.0        1140.0    114.0        1555.0       1074.0  11,9
54,0
3    172.0        1092.0    122.0        1584.0       1203.0  11,0
60,0
4    131.0        1205.0    116.0        1490.0       1110.0  11,2
59,6

       AH  Unnamed: 15  Unnamed: 16
0  0,7578          NaN          NaN
1  0,7255          NaN          NaN
2  0,7502          NaN          NaN
3  0,7867          NaN          NaN
4  0,7888          NaN          NaN
```

```python
df.shape
```

```
(9471, 17)
```

```python
#data cleaning
df=df.drop_duplicates()

df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 9358 entries, 0 to 9357
Data columns (total 17 columns):
 #   Column         Non-Null Count  Dtype  
---  ------         --------------  -----  
 0   Date           9357 non-null   object 
 1   Time           9357 non-null   object 
 2   CO(GT)         9357 non-null   object 
 3   PT08.S1(CO)    9357 non-null   float64
 4   NMHC(GT)       9357 non-null   float64
 5   C6H6(GT)       9357 non-null   object 
 6   PT08.S2(NMHC)  9357 non-null   float64
 7   NOx(GT)        9357 non-null   float64
 8   PT08.S3(NOx)   9357 non-null   float64
 9   NO2(GT)        9357 non-null   float64
 10  PT08.S4(NO2)   9357 non-null   float64
 11  PT08.S5(O3)    9357 non-null   float64
 12  T              9357 non-null   object 
 13  RH             9357 non-null   object 
 14  AH             9357 non-null   object 
 15  Unnamed: 15    0 non-null      float64
 16  Unnamed: 16    0 non-null      float64
dtypes: float64(10), object(7)
memory usage: 1.3+ MB
```

```
df.isna().sum()
```

```
Date                1
Time                1
CO(GT)              1
PT08.S1(CO)         1
NMHC(GT)            1
C6H6(GT)            1
PT08.S2(NMHC)       1
NOx(GT)             1
PT08.S3(NOx)        1
NO2(GT)             1
PT08.S4(NO2)        1
PT08.S5(O3)         1
T                   1
RH                  1
AH                  1
Unnamed: 15      9358
Unnamed: 16      9358
dtype: int64
```

```
df=df.drop(['Unnamed: 15','Unnamed: 16'],axis=1)
```

```
df
```

```
            Date       Time CO(GT)  PT08.S1(CO)  NMHC(GT) C6H6(GT)  \
0     10/03/2004  18.00.00    2,6       1360.0     150.0     11,9   
1     10/03/2004  19.00.00      2       1292.0     112.0      9,4   
2     10/03/2004  20.00.00    2,2       1402.0      88.0      9,0   
3     10/03/2004  21.00.00    2,2       1376.0      80.0      9,2   
4     10/03/2004  22.00.00    1,6       1272.0      51.0      6,5   
...          ...       ...    ...          ...       ...      ...   
9353  04/04/2005  11.00.00    2,4       1163.0    -200.0     11,4   
9354  04/04/2005  12.00.00    2,4       1142.0    -200.0     12,4   
9355  04/04/2005  13.00.00    2,1       1003.0    -200.0      9,5   
9356  04/04/2005  14.00.00    2,2       1071.0    -200.0     11,9   
9357         NaN       NaN    NaN          NaN       NaN      NaN   

      PT08.S2(NMHC)  NOx(GT)  PT08.S3(NOx)  NO2(GT)  PT08.S4(NO2)  \
0            1046.0    166.0        1056.0    113.0        1692.0   
1             955.0    103.0        1174.0     92.0        1559.0   
2             939.0    131.0        1140.0    114.0        1555.0   
3             948.0    172.0        1092.0    122.0        1584.0   
4             836.0    131.0        1205.0    116.0        1490.0   
...             ...      ...           ...      ...           ...   
9353         1027.0    353.0         604.0    179.0        1264.0   
9354         1063.0    293.0         603.0    175.0        1241.0   
9355          961.0    235.0         702.0    156.0        1041.0   
9356         1047.0    265.0         654.0    168.0        1129.0   
9357            NaN      NaN           NaN      NaN           NaN   

      PT08.S5(O3)     T    RH      AH  
0          1268.0  13,6  48,9  0,7578  
1           972.0  13,3  47,7  0,7255  
2          1074.0  11,9  54,0  0,7502  
3          1203.0  11,0  60,0  0,7867  
4          1110.0  11,2  59,6  0,7888  
...           ...   ...   ...     ...  
9353       1269.0  24,3  23,7  0,7119  
9354       1092.0  26,9  18,3  0,6406  
9355        770.0  28,3  13,5  0,5139  
9356        816.0  28,5  13,1  0,5028  
9357          NaN   NaN   NaN     NaN  

[9358 rows x 15 columns]
```

```python
df=df.rename(columns={'T':'Temperature'})
df=df.rename(columns={'AH':'Absolute Humidity'})
df=df.rename(columns={'RH':'Relative Humidity'})
df
```

```
            Date       Time CO(GT)  PT08.S1(CO)  NMHC(GT) C6H6(GT)  \
0     10/03/2004  18.00.00    2,6       1360.0     150.0     11,9   
1     10/03/2004  19.00.00      2       1292.0     112.0      9,4   
2     10/03/2004  20.00.00    2,2       1402.0      88.0      9,0   
```

```
3     10/03/2004  21.00.00    2,2       1376.0      80.0      9,2
4     10/03/2004  22.00.00    1,6       1272.0      51.0      6,5
...          ...       ...    ...          ...       ...      ...
9353  04/04/2005  11.00.00    2,4       1163.0    -200.0     11,4
9354  04/04/2005  12.00.00    2,4       1142.0    -200.0     12,4
9355  04/04/2005  13.00.00    2,1       1003.0    -200.0      9,5
9356  04/04/2005  14.00.00    2,2       1071.0    -200.0     11,9
9357         NaN       NaN    NaN          NaN       NaN      NaN

      PT08.S2(NMHC)  NOx(GT)  PT08.S3(NOx)  NO2(GT)  PT08.S4(NO2)  \
0            1046.0    166.0        1056.0    113.0        1692.0
1             955.0    103.0        1174.0     92.0        1559.0
2             939.0    131.0        1140.0    114.0        1555.0
3             948.0    172.0        1092.0    122.0        1584.0
4             836.0    131.0        1205.0    116.0        1490.0
...             ...      ...           ...      ...           ...
9353         1027.0    353.0         604.0    179.0        1264.0
9354         1063.0    293.0         603.0    175.0        1241.0
9355          961.0    235.0         702.0    156.0        1041.0
9356         1047.0    265.0         654.0    168.0        1129.0
9357            NaN      NaN           NaN      NaN           NaN

      PT08.S5(O3) Temperature Relative Humidity Absolute Humidity
0          1268.0        13,6              48,9            0,7578
1           972.0        13,3              47,7            0,7255
2          1074.0        11,9              54,0            0,7502
3          1203.0        11,0              60,0            0,7867
4          1110.0        11,2              59,6            0,7888
...           ...         ...               ...               ...
9353       1269.0        24,3              23,7            0,7119
9354       1092.0        26,9              18,3            0,6406
9355        770.0        28,3              13,5            0,5139
9356        816.0        28,5              13,1            0,5028
9357          NaN         NaN               NaN               NaN

[9358 rows x 15 columns]
```

```
df=df.replace(',','.',regex=True)

float_col=["CO(GT)","C6H6(GT)","Temperature","Relative
Humidity","Absolute Humidity"]
df[float_col]=df[float_col].astype(float)

df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 9358 entries, 0 to 9357
Data columns (total 15 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
```

```
 0   Date                  9357 non-null   object
 1   Time                  9357 non-null   object
 2   CO(GT)                9357 non-null   float64
 3   PT08.S1(CO)           9357 non-null   float64
 4   NMHC(GT)              9357 non-null   float64
 5   C6H6(GT)              9357 non-null   float64
 6   PT08.S2(NMHC)         9357 non-null   float64
 7   NOx(GT)               9357 non-null   float64
 8   PT08.S3(NOx)          9357 non-null   float64
 9   NO2(GT)               9357 non-null   float64
 10  PT08.S4(NO2)          9357 non-null   float64
 11  PT08.S5(O3)           9357 non-null   float64
 12  Temperature           9357 non-null   float64
 13  Relative Humidity     9357 non-null   float64
 14  Absolute Humidity     9357 non-null   float64
dtypes: float64(13), object(2)
memory usage: 1.1+ MB
```

```
df=df.drop(['Date','Time'],axis=1)
```

```
df.isna().sum()
```

```
CO(GT)                    1
PT08.S1(CO)               1
NMHC(GT)                  1
C6H6(GT)                  1
PT08.S2(NMHC)             1
NOx(GT)                   1
PT08.S3(NOx)              1
NO2(GT)                   1
PT08.S4(NO2)              1
PT08.S5(O3)               1
Temperature               1
Relative Humidity         1
Absolute Humidity         1
dtype: int64
```

```
df=df.dropna()
```

```
df.isna().sum()
```

```
CO(GT)                    0
PT08.S1(CO)               0
NMHC(GT)                  0
C6H6(GT)                  0
PT08.S2(NMHC)             0
NOx(GT)                   0
PT08.S3(NOx)              0
NO2(GT)                   0
PT08.S4(NO2)              0
PT08.S5(O3)               0
```

```
Temperature             0
Relative Humidity       0
Absolute Humidity       0
dtype: int64

df.head()

   CO(GT)  PT08.S1(CO)  NMHC(GT)  C6H6(GT)  PT08.S2(NMHC)  NOx(GT)  \
0     2.6       1360.0     150.0      11.9         1046.0    166.0
1     2.0       1292.0     112.0       9.4          955.0    103.0
2     2.2       1402.0      88.0       9.0          939.0    131.0
3     2.2       1376.0      80.0       9.2          948.0    172.0
4     1.6       1272.0      51.0       6.5          836.0    131.0

   PT08.S3(NOx)  NO2(GT)  PT08.S4(NO2)  PT08.S5(O3)  Temperature  \
0        1056.0    113.0        1692.0       1268.0         13.6
1        1174.0     92.0        1559.0        972.0         13.3
2        1140.0    114.0        1555.0       1074.0         11.9
3        1092.0    122.0        1584.0       1203.0         11.0
4        1205.0    116.0        1490.0       1110.0         11.2

   Relative Humidity  Absolute Humidity
0               48.9             0.7578
1               47.7             0.7255
2               54.0             0.7502
3               60.0             0.7867
4               59.6             0.7888

#data integration
common_col=['CO(GT)','NO2(GT)']
data1=df[common_col+['C6H6(GT)','NOx(GT)']]
data2=df[common_col+
["PT08.S1(CO)","PT08.S2(NMHC)","PT08.S3(NOx)","PT08.S4(NO2)","PT08.S5(
O3)","Temperature","Relative Humidity","Absolute Humidity"]]

data1_s=data1.head(100)
data2_s=data2.head(100)

inner_merged=pd.merge(data1_s,data2_s,on=common_col,how="inner")

inner_merged.head()

   CO(GT)  NO2(GT)  C6H6(GT)  NOx(GT)  PT08.S1(CO)  PT08.S2(NMHC)  \
0     2.6    113.0      11.9    166.0       1360.0         1046.0
1     2.0     92.0       9.4    103.0       1292.0          955.0
2     2.2    114.0       9.0    131.0       1402.0          939.0
3     2.2    122.0       9.2    172.0       1376.0          948.0
4     1.6    116.0       6.5    131.0       1272.0          836.0

   PT08.S3(NOx)  PT08.S4(NO2)  PT08.S5(O3)  Temperature  Relative
Humidity  \
```

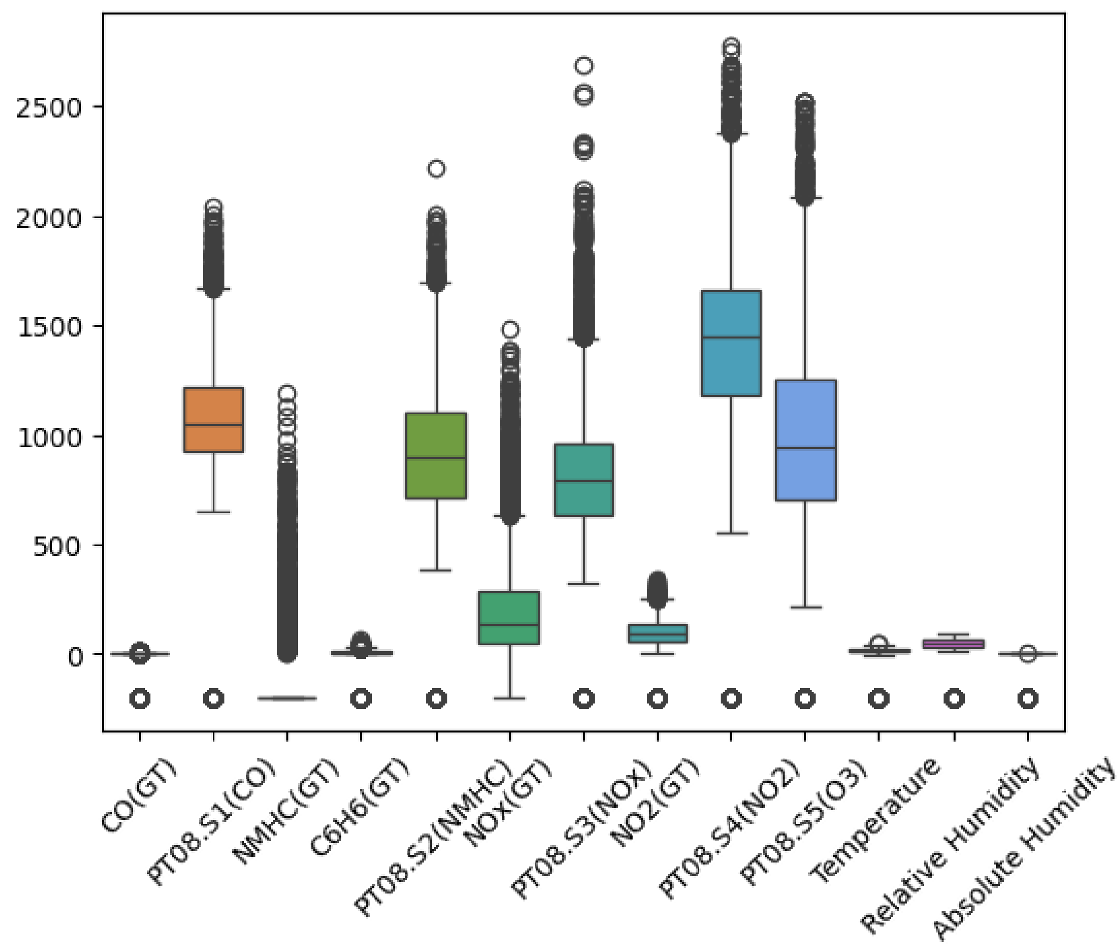
```
0       1056.0        1692.0       1268.0          13.6
48.9
1       1174.0        1559.0        972.0          13.3
47.7
2       1140.0        1555.0       1074.0          11.9
54.0
3       1092.0        1584.0       1203.0          11.0
60.0
4       1205.0        1490.0       1110.0          11.2
59.6

   Absolute Humidity
0             0.7578
1             0.7255
2             0.7502
3             0.7867
4             0.7888
```

```
left_merged=pd.merge(data1_s,data2_s,on=common_col,how='left')
left_merged.head()
```

```
   CO(GT)  NO2(GT)  C6H6(GT)  NOx(GT)  PT08.S1(CO)  PT08.S2(NMHC)  \
0     2.6    113.0      11.9    166.0       1360.0         1046.0
1     2.0     92.0       9.4    103.0       1292.0          955.0
2     2.2    114.0       9.0    131.0       1402.0          939.0
3     2.2    122.0       9.2    172.0       1376.0          948.0
4     1.6    116.0       6.5    131.0       1272.0          836.0

   PT08.S3(NOx)  PT08.S4(NO2)  PT08.S5(O3)  Temperature  Relative
Humidity  \
0       1056.0        1692.0       1268.0          13.6
48.9
1       1174.0        1559.0        972.0          13.3
47.7
2       1140.0        1555.0       1074.0          11.9
54.0
3       1092.0        1584.0       1203.0          11.0
60.0
4       1205.0        1490.0       1110.0          11.2
59.6

   Absolute Humidity
0             0.7578
1             0.7255
2             0.7502
3             0.7867
4             0.7888
```

```python
#error correction
sns.boxplot(df)
plt.xticks(rotation=45)
```

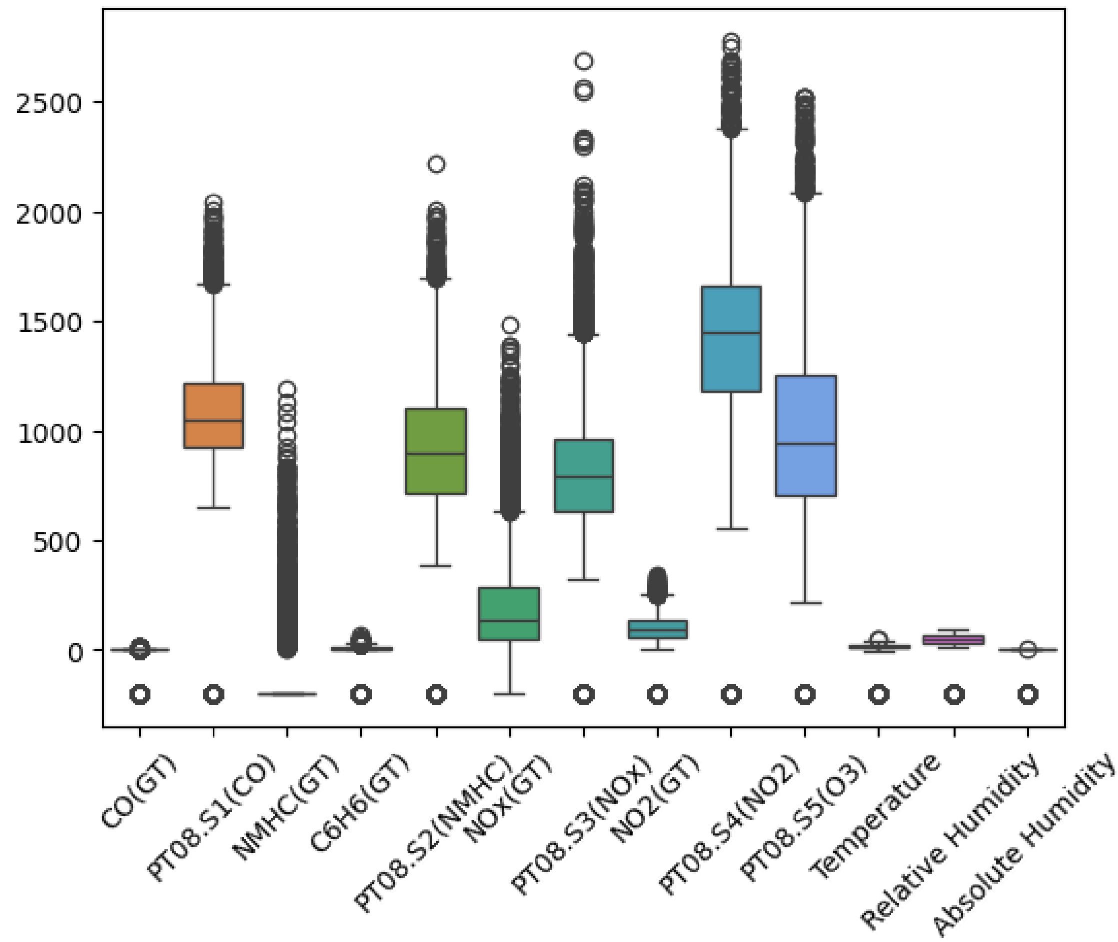
```
([0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12],
 [Text(0, 0, 'CO(GT)'),
  Text(1, 0, 'PT08.S1(CO)'),
  Text(2, 0, 'NMHC(GT)'),
  Text(3, 0, 'C6H6(GT)'),
  Text(4, 0, 'PT08.S2(NMHC)'),
  Text(5, 0, 'NOx(GT)'),
  Text(6, 0, 'PT08.S3(NOx)'),
  Text(7, 0, 'NO2(GT)'),
  Text(8, 0, 'PT08.S4(NO2)'),
  Text(9, 0, 'PT08.S5(O3)'),
  Text(10, 0, 'Temperature'),
  Text(11, 0, 'Relative Humidity'),
  Text(12, 0, 'Absolute Humidity')])
```



```
def remove_outliers(col):
    Q1=col.quantile(0.25)
    Q3=col.quantile(0.75)
    IQR=Q3-Q1
    threshold=2.5*IQR
    outlier_mask=(col<Q1-threshold) | (col>Q3+threshold)
    return col[~outlier_mask]

df.columns
```

```
Index(['CO(GT)', 'PT08.S1(CO)', 'NMHC(GT)', 'C6H6(GT)',
'PT08.S2(NMHC)',
       'NOx(GT)', 'PT08.S3(NOx)', 'NO2(GT)', 'PT08.S4(NO2)',
'PT08.S5(O3)',
       'Temperature', 'Relative Humidity', 'Absolute Humidity'],
      dtype='object')
```

```
num_cols=df.select_dtypes(include=['number']).columns

for col in num_cols:
    cleaned_col=remove_outliers(df[col])
    df.loc[cleaned_col.index,col]=cleaned_col

sns.boxplot(df)
plt.xticks(rotation=45)
plt.show()
```




```
from sklearn.preprocessing import MinMaxScaler

scaler=MinMaxScaler()
df[num_cols]=scaler.fit_transform(df[num_cols])
df.head()
```

```
     CO(GT)  PT08.S1(CO)  NMHC(GT)  C6H6(GT)  PT08.S2(NMHC)   NOx(GT)  \
0  0.956111     0.696429  0.251980  0.803565       0.516156  0.217987
1  0.953280     0.666071  0.224622  0.794084       0.478459  0.180465
2  0.954224     0.715179  0.207343  0.792567       0.471831  0.197141
3  0.954224     0.703571  0.201584  0.793326       0.475559  0.221560
4  0.951392     0.657143  0.180706  0.783087       0.429163  0.197141


   PT08.S3(NOx)   NO2(GT)  PT08.S4(NO2)  PT08.S5(O3)  Temperature  \
0      0.435657  0.579630      0.635966     0.539111     0.873262
```

```
1      0.476587  0.540741      0.591261     0.430408     0.872036
2      0.464794  0.581481      0.589916     0.467866     0.866312
3      0.448144  0.596296      0.599664     0.515241     0.862633
4      0.487340  0.585185      0.568067     0.481087     0.863451

   Relative Humidity  Absolute Humidity
0           0.862141           0.992715
1           0.857984           0.992556
2           0.879806           0.992678
3           0.900589           0.992858
4           0.899203           0.992869
```

```
#data model building
y=df['Temperature']
X=df.drop('Temperature',axis=1)

from sklearn.model_selection import train_test_split
from sklearn.metrics import
r2_score,mean_absolute_error,mean_squared_error
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,rando
m_state=42)

from sklearn.linear_model import LinearRegression
model1=LinearRegression()
model1.fit(X_train,y_train)
y_pred=model1.predict(X_test)

print("r2 score:",r2_score(y_test,y_pred))
print("mean absolute error:",mean_absolute_error(y_test,y_pred))
print("mean squared error:",mean_squared_error(y_test,y_pred))
print("accuracy:",r2_score(y_test,y_pred)*100)
```

```
r2 score: 0.995077596394832
mean absolute error: 0.009681121939987306
mean squared error: 0.00015160045647323898
accuracy: 99.5077596394832
```

```
from sklearn.tree import DecisionTreeRegressor
model2=DecisionTreeRegressor()
model2.fit(X_train,y_train)
y_pred=model2.predict(X_test)

print("r2 score:",r2_score(y_test,y_pred))
print("mean absolute error:",mean_absolute_error(y_test,y_pred))
print("mean squared error:",mean_squared_error(y_test,y_pred))
print("accuracy:",r2_score(y_test,y_pred)*100)
```

```
r2 score: 0.9998792916282065
mean absolute error: 0.001236319894332979
mean squared error: 3.717583061418078e-06
accuracy: 99.98792916282065
```

```
from sklearn.ensemble import RandomForestRegressor
model3 = RandomForestRegressor()

from sklearn.model_selection import cross_val_score
scores = cross_val_score(model3, X, y, cv=5, scoring='r2')
print("Mean R2 from Cross-validation:", scores.mean())
```